Learning to Ask for More Information: Progressive Reveal and Chimera for Abstention

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Abstract

Multiple-choice language models are usually evaluated in a forced-answer setting, where they must choose A/B/C/D even when the input is incomplete. This project studies whether a small Qwen model can learn to abstain when only partial question information is visible and answer more often as additional text is revealed. The main method is a **truncation-based abstention objective** with an explicit DK ("do not know") option. Because truncation-only training encouraged a shortcut in which short or broken prompts implied DK, we add **Chimera** questions: full-length, fluent questions that are synthetically made unanswerable and labeled with soft DK targets. We compare the resulting five-way A/B/C/D/DK model against a regular-model confidence-threshold baseline on MMLU-style evaluation and test transfer on SciQ, GSM8K, and MMLU-Pro. The model learns a meaningful abstention signal, but it does not consistently outperform maximum-softmax confidence on the main evaluation. SciQ shows useful abstention in absolute terms, while GSM8K and MMLU-Pro remain weak or inconclusive under our four-choice conversions.

1 Introduction

Language models often answer even when they are uncertain. For multiple-choice question answering this is especially visible: a model must choose A/B/C/D, so forced-answer accuracy rewards both knowledge and lucky guessing. The motivating analogy in this project is simple: when a person cannot see the full question, they can only guess; when more of the question becomes visible, the chance of answering correctly should increase. A useful model should therefore be able to say "I do not know" when the visible context is insufficient, and answer only when enough information has been revealed. This project studies this behavior through *selective prediction*: a model can either answer or abstain. The goal is not only to improve the accuracy of answered examples, but also to illustrate whether a model can learn the pattern "insufficient information $\rightarrow$ ask for more information".

The research question is:

> *Can a small multiple-choice language model learn, from truncated questions, to abstain when the visible information is insufficient and answer more often as additional question text is revealed?*

We focus on a small Qwen model trained for four-choice multiple-choice QA. The baseline model predicts only A/B/C/D and abstains only by an external threshold on maximum softmax confidence. Our proposed model extends the output space to A/B/C/D/DK and trains the DK class directly. The central method is truncation: questions are partially revealed, and early partial versions receive more DK supervision than full questions. The conceptual hope is that the model will learn a useful interaction pattern: when it predicts DK, more text should be revealed, and only then should the model become more willing to answer.

A key empirical challenge emerged during development. Training only on truncated questions encouraged a superficial shortcut: shorter or logically broken prompts were often associated with DK labels. To separate abstention from input length, we introduced **Chimera** examples: full-length questions whose answer options are synthetically corrupted so that abstention is appropriate. All final results therefore compare the chimera/truncation five-way model against the regular confidence baseline.

Our findings are deliberately conservative. The explicit DK model learns a real uncertainty signal, but ordinary confidence thresholding remains very strong and is usually better at the same coverage on the main MMLU-style curves. Cross-dataset results are also mixed: SciQ is positive in absolute terms, while GSM8K and MMLU-Pro should be interpreted cautiously. Thus, the project is both a proof of concept and useful negative evidence: explicit DK supervision is promising, but the current signal is not enough to consistently beat standard confidence-based abstention.

2 Related Work

Selective prediction. Selective prediction allows a classifier to abstain from low-confidence inputs. The usual evaluation studies the trade-off between coverage and risk, where risk is the error rate among answered examples. Xin et al. (2021) study selective prediction in NLP and compare confidence estimators and error regularization. Our work follows the same evaluation philosophy but studies a task-specific explicit abstention class for multiple-choice LMs.

Calibration and confidence. Neural probabilities are often miscalibrated. Guo et al. (2017) show that modern neural networks can be overconfident and that temperature scaling can improve calibration. For this reason we evaluate not only accuracy, but also AUROC and expected calibration error (ECE). The baseline in our project is maximum-softmax confidence, because it is a simple and strong confidence estimator for selective prediction.

Language models and self-knowledge. Several works ask whether LMs can estimate their own correctness. Kadavath et al. (2022) study whether LMs know what they know, including self-evaluation probabilities such as $P(\mathrm{True})$ and $P(\mathrm{IK})$. Manakul et al. (2023) propose SelfCheckGPT, which detects hallucination by checking consistency across sampled responses. Our setup is smaller and more controlled: we train a small model to output an explicit DK option in multiple-choice QA.

Benchmarks. MMLU evaluates broad multitask language understanding across many subjects (Hendrycks et al., 2021). MMLU-Pro extends MMLU by making the benchmark more reasoning-oriented and expanding the answer set to ten options (Wang et al., 2024). In this project, MMLU-Pro is converted to four options for compatibility with the trained model, which is a limitation. We also evaluate on SciQ and a four-choice conversion of GSM8K. GSM8K is originally open-ended, so our GSM8K numbers should be interpreted only as a stress test of the abstention mechanism, not as canonical GSM8K performance.

3 Method

3.1 Task Formulation

Each original example contains a question x, four choices A, B, C, D, and a gold label $y \in \{A, B, C, D\}$. A standard model computes answer probabilities

$$p_\theta(c \mid x) = \mathrm{softmax}(z_A, z_B, z_C, z_D)_c.$$

The baseline predicts the most likely answer and uses an external threshold on

$$s_{\text{base}} = \max_{c \in \{A,B,C,D\}} p_\theta(c \mid x)$$

to decide whether to answer.

Our Chimera method changes the output space to five classes:

$$\{A, B, C, D, \mathrm{DK}\}.$$

The model produces a five-way distribution, and $p(\mathrm{DK})$ is interpreted as an abstention signal. At inference we test both direct DK thresholding and derived confidence scores such as

$$s_{\text{ans}} = \max_{c \in \{A,B,C,D\}} p(c), \tag{1}$$

$$s_{\text{notDK}} = 1 - p(\mathrm{DK}), \tag{2}$$

$$s_{\text{prod}} = s_{\text{ans}}(1 - p(\mathrm{DK})), \tag{3}$$

$$s_{\text{margin}} = s_{\text{ans}} - p(\mathrm{DK}). \tag{4}$$

For a pure DK threshold, the rule is

$$\text{answer if } p(\mathrm{DK}) < \tau, \quad \text{otherwise abstain.}$$

Thus, for this rule, a larger τ produces higher coverage.

3.2 Truncation Supervision: the Main Idea

The main source of DK supervision is truncation. Given a full question, we construct partial versions with reveal ratios such as $1/m, 2/m, \ldots, 1$. In the final reported experiments we use $m = 3$, because among the explored settings it gave the best validation behavior. We also experimented with an attention-based reveal strategy, but it performed worse than the simple progressive split used in the final system. Therefore, the reported method uses the normal progressive reveal construction: early steps contain a smaller prefix of the question, and later steps contain more of the same question. The intended behavior is analogous to a human reading only part of a question: at early reveal ratios, there may not be enough information to answer, so the model should prefer DK; as the reveal ratio grows, the model should become more willing to answer. In an interactive interpretation, DK means "show me more of the question."

We use soft labels rather than only hard labels. For a reveal ratio r, the target approximately assigns

$$q_y = r, \quad q_{\mathrm{DK}} = 1 - r,$$

with smoothing/clipping in the implementation. A full question therefore mostly trains the correct answer, while an early truncated question gives more mass to DK. This design tries to teach a gradual transition from abstention to answering, rather than a binary rule.

However, truncation-only training performed poorly. Our interpretation is that it created an unintended shortcut: the model could learn that short prompts, incomplete sentences, or logically broken text imply DK. This explains why truncation alone did not reliably improve selective prediction and motivated the chimera correction.

3.3 Chimera Questions

Chimera is an auxiliary component added after observing the truncation shortcut. Each chimera keeps the input full-length and fluent, but corrupts the answer options so that none of the displayed choices should be trusted. This gives the model DK supervision that is not explained by shortness alone.

Let the original example be (x, C, y), where x is the question, $C = (c_A, c_B, c_C, c_D)$ are the choices, and y is the correct answer index. For every clean example, the dataset adds the original question with a one-hot answer target. Then, with probability ρ, it attempts to add one chimera example; in the final run $\rho = 1$. If no valid replacement is found, the chimera example is skipped.

The implementation uses two mechanisms. **Remove-correct-answer Chimera** keeps the original question and wrong choices, but replaces the correct choice with a wrong option sampled from another question, preferably from the same MMLU subject. **Same-subject choice-swap Chimera** keeps the original question but replaces all four choices with choices from another same-subject question. In both cases, the question text itself is not mixed with another question; only the answer choices are changed. Same-subject sampling makes the corrupted options plausible, reducing the chance that the model solves the task by detecting an obvious topic mismatch.

The effective chimera hyperparameters are: the amount ρ (`chimera_ratio`), the mixture proportion π between the two mechanisms, and the soft DK target δ (`dk_prob`). The final implementation uses an approximately balanced 50/50 mixture between the two chimera types. For chimera examples, the target distribution is

$$q_{\mathrm{DK}} = \delta, \qquad q_A = q_B = q_C = q_D = \frac{1-\delta}{4}.$$

Thus the model is encouraged, but not forced, to put most probability mass on DK. In the strongest run, the best coverage-based behavior came from adding all possible chimera examples and using $\delta \approx 0.5$, which should be viewed as a tuned setting rather than a general optimum.

3.4 Training Objective

The model is trained with cross entropy against a soft target distribution over the five labels. For each example with target distribution q, the loss is

$$\mathcal{L}(x, q) = - \sum_{k \in \{A,B,C,D,\mathrm{DK}\}} q_k \log p_\theta(k \mid x).$$

We explicitly construct three dataset variants during development and training. The first is the **regular dataset**, containing the original full questions with one-hot A/B/C/D targets. The second is the **truncation dataset**, containing partially revealed versions of questions with soft targets that shift mass between the correct answer and DK according to the reveal ratio. The third is the **Chimera dataset**, containing full-length questions with corrupted answer options and soft DK targets. The final reported model uses this five-way chimera/truncation setup. In the main training run, we sampled up to 100,000 original training questions, yielding 99,842 available MMLU auxiliary-training examples. After constructing the regular, truncated, and Chimera variants, this produced an effective training set of about 600,000 training instances. This number should be understood as the expanded training-instance count, not as 600,000 independent original questions.

One implementation choice was specific to the truncation-training stage: we kept the truncated variants in their constructed progressive order instead of shuffling them freely. This was done to preserve the reveal structure of each original question: the model sees related variants of the same question from less information to more information. Thus, the no-shuffle choice is an explicit curriculum-style design for the progressive-reveal data, not a dependence on random shuffling. The regular training dataset and the Chimera training dataset are still shuffled, and the train/validation/test splits remain separate. Future work should compare this simple grouped reveal curriculum with other explicit curricula, such as mixing reveal levels or sampling reveal steps independently.

4 Experimental Design

4.1 Models

We compare two main systems:

1. **Baseline**: a regular four-choice model with maximum-softmax confidence thresholding.

2. **Chimera**: a five-way A/B/C/D/DK model trained with truncation and chimera examples.

All plots discussed in this report correspond to these two systems.

Both systems are based on Qwen3-0.6B (Yang et al., 2025) and are fine-tuned using LoRA rather than full-parameter fine-tuning (Hu et al., 2022). The reported LoRA configuration uses rank $r = 16$, $\alpha = 16$, dropout 0.05, no bias adaptation, and targets the attention and MLP projection modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. We used LoRA to make the experiment feasible on limited GPU resources while still allowing the model to adapt its answer and DK behavior. The maximum token length was set to 512 to reduce runtime and memory use; for most MMLU-style questions this retains the full prompt and all four answer choices.

4.2 Datasets

The main evaluation uses an MMLU-style multiple-choice setup. Cross-dataset evaluation uses:

- **SciQ**: science multiple-choice questions.

- **GSM8K**: math word problems converted into artificial four-choice numeric questions.

- **MMLU-Pro**: converted into four choices for compatibility.

The conversions for GSM8K and MMLU-Pro are not standard benchmark protocols. They are used only to test transfer of the abstention behavior.

4.3 Threshold Selection

For coverage-based experiments, thresholds are selected on validation data. For example, to target 70% coverage, we find the validation threshold that answers approximately 70% of validation examples, then apply that threshold once to the test set. This prevents choosing thresholds directly on the test set.

For the baseline, the threshold is on maximum answer confidence. For the DK model, we evaluate several scores. When using pure $p(\mathrm{DK})$, the threshold rule is inverted relative to confidence: low $p(\mathrm{DK})$ means answer, high $p(\mathrm{DK})$ means abstain.

4.4 Metrics

We report:

- **Forced accuracy**: accuracy when the model must answer A/B/C/D.

- **Coverage**: fraction of examples answered.

- **Selective accuracy**: accuracy among answered examples.

- **DK rate**: fraction abstained.

- **Effective accuracy**: fraction of all examples that are answered correctly.

- **AUROC**: how well a confidence score ranks forced-correct examples above forced-wrong examples.

- **ECE**: expected calibration error.

- **Risk-coverage**: risk is $1-$selective accuracy; lower is better.

Implementation sanity. The notebook also records training-loss curves for the baseline and DK stages as a basic implementation sanity check: the optimization loop, label mapping, and five-way output head should produce decreasing training loss before the final evaluation is trusted. This is not treated as a performance result. A stronger future check would report a full small-subset overfitting curve, but the existing loss tracking helped verify that the training pipeline was connected correctly.

Method	Forced Acc.	Coverage	Sel. Acc.	AUROC	ECE
Baseline conf. threshold	0.487	0.706	0.562	0.714	0.066
DK Chimera argmax	0.493	0.980	0.497	0.694	0.129

Table 1: Main summary metrics from the final MMLU-style run. The baseline row uses a confidence threshold selected on validation for approximately 70% target coverage. The DK argmax row answers when the largest probability is A/B/C/D and abstains only when the largest probability is DK.

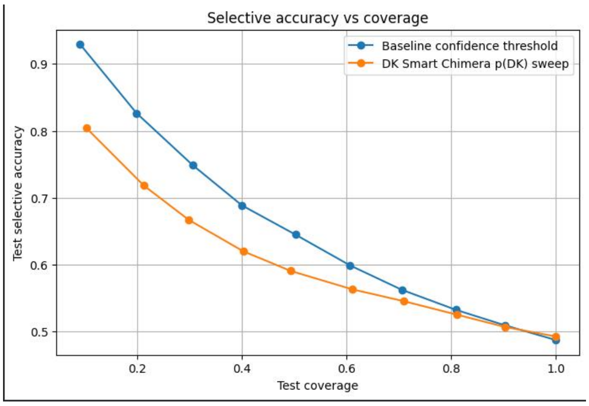

Figure 1: Main MMLU-style selective accuracy versus coverage. The regular confidence baseline remains above the direct Chimera $p(\mathrm{DK})$ sweep across the curve.

4.5 Code Availability

The project code and Colab notebook are available at: `https://github.com/DavidZago/Progressive-Reveal-and-Chimera-for-Abstention`

5 Results

This section reports the exact outputs exported from the final notebook run. Unless stated otherwise, coverage thresholds are selected on the validation split and then evaluated once on the test split. The main MMLU-style test set contains 14,042 examples.

5.1 Main MMLU-Style Evaluation

Table 1 shows the basic operating points. The regular confidence baseline has forced accuracy 0.487 and, after selecting a validation threshold for 70% target coverage, reaches 0.706 test coverage and 0.562 selective accuracy. The argmax-only DK rule abstains very rarely: it keeps 0.980 coverage and reaches only 0.497 selective accuracy. Therefore, argmax-only DK is not a useful decision rule by itself in this run. The more informative evaluation is the full coverage sweep.

Target cov.	Base test cov.	Base sel. acc.	DK test cov.	DK sel. acc.
1.0	1.000	0.487	1.000	0.493
0.9	0.904	0.509	0.904	0.507
0.8	0.809	0.532	0.811	0.525
0.7	0.706	0.562	0.710	0.545
0.6	0.607	0.599	0.611	0.563
0.5	0.502	0.645	0.493	0.591
0.4	0.400	0.689	0.402	0.620
0.3	0.305	0.749	0.298	0.667
0.2	0.198	0.826	0.212	0.718
0.1	0.090	0.929	0.102	0.804

Table 2: Selective accuracy at validation-selected target coverages. The baseline is maximum-softmax confidence. The DK column is the direct $p(\mathrm{DK})$ sweep, where examples with lower $p(\mathrm{DK})$ are answered. Higher selective accuracy is better.

Score	AUROC ↑	ECE ↓
Baseline max confidence	0.714	0.066
Baseline entropy confidence	0.709	0.226
DK entropy confidence	0.696	0.108
DK conditional answer confidence	0.695	0.143
DK product confidence	0.694	0.129
DK raw answer confidence	0.694	0.129
DK max-answer minus $p(\mathrm{DK})$	0.693	–
DK $1 - p(\mathrm{DK})$	0.643	0.471

Table 3: Score quality on the main MMLU-style test set. AUROC measures whether the score ranks forced-correct examples above forced-wrong examples. ECE measures calibration error.

Table 2 and Figure 1 are the main result. The Chimera $p(\mathrm{DK})$ signal is useful: selective accuracy rises from 0.493 at full coverage to 0.804 at roughly 10% coverage. However, the ordinary confidence baseline is stronger at every listed coverage, including the central 70% point (0.562 versus 0.545 selective accuracy) and the low-coverage region (0.929 versus 0.804 at roughly 10% coverage).

5.2 Which DK Score Works Best?

Table 3 explains why the baseline remains hard to beat. The best baseline score reaches AUROC 0.714 and ECE 0.066. DK-derived scores are close but lower: DK entropy confidence reaches AUROC 0.696 and ECE 0.108, while direct $1 - p(\mathrm{DK})$ is weaker with AUROC 0.643 and ECE 0.471. Thus, $p(\mathrm{DK})$ is meaningful, but it is not as well calibrated as regular answer confidence.

Table 4 shows that abstention is useful for both systems: wrong examples are rejected more often than correct examples. The best DK-derived scores are close to the baseline, but still do not clearly improve it. For example, DK conditional confidence

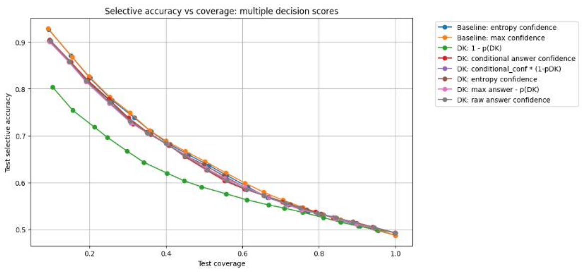

Figure 2: Main MMLU-style comparison across several decision scores. DK answer-confidence based scores are close to the baseline, while direct $1-p(\mathrm{DK})$ is weaker.

Method	Cov.	Sel. Acc.	Wrong rej.	Correct rej.
Baseline max confidence	0.706	0.562	0.398	0.186
DK argmax-only	0.980	0.497	0.028	0.011
DK raw answer conf.	0.712	0.556	0.377	0.197
DK $1-p(\mathrm{DK})$	0.710	0.545	0.364	0.214
DK margin	0.714	0.553	0.370	0.198
DK conditional conf.	0.704	0.559	0.388	0.201
DK product conf.	0.712	0.556	0.377	0.197

Table 4: Rejection quality around the 70% target coverage setting. "Wrong rej." is the fraction of forced-wrong examples rejected; "Correct rej." is the fraction of forced-correct examples rejected.

reaches 0.559 selective accuracy at 0.704 coverage, compared with 0.562 at 0.706 coverage for the baseline. This is an important honest result: the explicit DK model is competitive, but the simple baseline is slightly better on the main evaluation.

5.3 Truncation Reveal Analysis

Table 5 supports the main motivation of the project. As the reveal ratio increases from one third to the full question, average $p(\mathrm{DK})$ drops from 0.244 to 0.037, and argmax coverage rises from 0.602 to 0.980. This is the desired qualitative pattern: the model learned that heavily truncated questions should receive more DK probability, while fuller questions should be answered more often. The quantitative accuracy gain is modest but consistent: selective accuracy rises from 0.406 to 0.497.

A useful implementation finding was that chimera details mattered. The best coverage-based behavior among the explored chimera settings appeared when the training set included as many chimera examples as possible, the two chimera mechanisms were used in an approximately balanced ratio, and the chimera soft label assigned about half of the target mass to DK. This supports the intuition behind Chimera: the model benefits from seeing full-length unanswerable examples, but overly hard DK targets can make the abstention signal too aggressive.

Step	Reveal	Avg. $p(\mathrm{DK})$	Forced Acc.	Coverage	Sel. Acc.
1	0.333	0.244	0.375	0.602	0.406
2	0.667	0.151	0.429	0.794	0.457
3	1.000	0.037	0.493	0.980	0.497

Table 5: Progressive-reveal diagnostic on the test set. Unlike the main full-question evaluation, this analysis evaluates each test question at three reveal levels: one-third, two-thirds, and the full question. The DK argmax rule answers when the largest probability is one of A–D and abstains when the largest probability is DK. As more of the question is visible, average $p(\mathrm{DK})$ decreases and coverage increases.

5.4 Cross-Dataset Evaluation

Table 6 and Figure 3 give the clearest cross-dataset picture. SciQ is the strongest positive transfer case in absolute terms. The baseline reaches 0.941 selective accuracy at about 0.699 coverage. DK-derived scores are also useful: DK answer confidence reaches 0.917 selective accuracy at 0.701 coverage, and DK product confidence reaches 0.916 at 0.705 coverage. However, even on SciQ the baseline remains stronger.

GSM8K is a useful but negative stress test. In our artificial four-choice numeric conversion, absolute performance is low: forced accuracy is 0.186 for the baseline and 0.234 for the DK model. The DK curves are above the baseline in several coverage ranges, especially when using scores related to $1-p(\mathrm{DK})$. However, the important point is that the best selective accuracies remain around 0.20–0.23, below the 25% random-choice level of a balanced four-choice task. Therefore, we do not treat GSM8K as a successful result. At most, it suggests that the DK-trained model changes the ranking of examples relative to the baseline under this artificial conversion, but the converted task itself is not being solved.

MMLU-Pro is mixed, with one important positive low-coverage behavior. At the beginning of the coverage curve, where the systems answer only the examples they consider most reliable, DK-based scores are clearly above the baseline. For example, in the lowest-coverage region the baseline is around 0.60 selective accuracy, while the best DK curves reach roughly 0.65–0.68, an improvement of about 5–8 absolute percentage points and roughly 10% relative improvement. However, this advantage is local: at the 70% target point the baseline reaches

Dataset	Method / score	Forced Acc.	AUROC	ECE	Cov. @0.7	Sel. Acc. @0.7
SciQ	Baseline max confidence	0.824	0.861	0.041	0.699	0.941
SciQ	DK $1 - p(\mathrm{DK})$	0.802	0.664	0.181	0.703	0.835
SciQ	DK answer confidence	0.802	0.828	0.050	0.701	0.917
SciQ	DK product	0.802	0.820	0.052	0.705	0.916
SciQ	DK margin	0.802	0.818	0.053	0.703	0.917
SciQ	DK argmax-only	0.802	–	–	0.991	0.804
GSM8K	Baseline max confidence	0.186	0.350	0.168	0.740	0.155
GSM8K	DK $1 - p(\mathrm{DK})$	0.234	0.447	0.720	0.647	0.210
GSM8K	DK answer confidence	0.234	0.406	0.161	0.667	0.202
GSM8K	DK product	0.234	0.406	0.154	0.670	0.207
GSM8K	DK margin	0.234	0.409	0.155	0.666	0.210
GSM8K	DK argmax-only	0.234	–	–	0.995	0.234
MMLU-Pro	Baseline max confidence	0.395	0.586	0.052	0.442	0.482
MMLU-Pro	DK $1 - p(\mathrm{DK})$	0.403	0.534	0.548	0.710	0.413
MMLU-Pro	DK answer confidence	0.403	0.596	0.071	0.569	0.457
MMLU-Pro	DK product	0.403	0.591	0.086	0.595	0.455
MMLU-Pro	DK margin	0.403	0.585	0.100	0.626	0.444
MMLU-Pro	DK argmax-only	0.403	–	–	0.975	0.409

Table 6: Cross-dataset evaluation on SciQ, GSM8K, and MMLU-Pro. "Cov. @0.7" and "Sel. Acc. @0.7" report the test coverage and selective accuracy obtained after choosing a threshold on validation for approximately 70% target coverage. GSM8K and MMLU-Pro are converted to four-choice format and should not be interpreted as standard benchmark results.

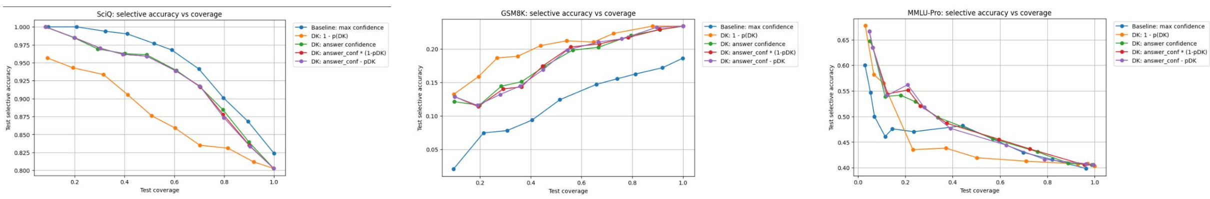

Figure 3: Cross-dataset selective accuracy versus coverage. SciQ is strong in absolute accuracy, but the baseline remains stronger. GSM8K shows DK curves above the baseline in several regions, but all curves remain below random four-choice accuracy, so this is not a successful GSM8K result. MMLU-Pro is mixed: DK-based scores improve over the baseline at the earliest, lowest-coverage operating points, but the advantage does not hold consistently at medium and high coverage.

0.482 selective accuracy at 0.442 coverage, while DK answer confidence reaches 0.457 at 0.569 coverage. Direct $1 - p(\mathrm{DK})$ also has poor calibration, with ECE 0.548. Therefore, MMLU-Pro should be described as evidence that Chimera can improve the very strict low-coverage regime, but not as a robust win over the baseline across the whole curve. Because MMLU-Pro was converted from ten choices to four, these results should be treated as a transfer stress test rather than a benchmark claim.

6 Discussion

The central lesson is that explicit DK supervision is useful but not sufficient. The Chimera model learns a non-random abstention signal: wrong examples often have higher $p(\mathrm{DK})$, and rejected examples are enriched for mistakes. However, ordinary answer confidence already captures much of the same information, so the best DK-derived curves are close to the baseline rather than clearly better.

The truncation-only failure exposed the main difficulty: the desired behavior is "I need more information," not "the input is short." Chimera improved the training signal by adding full-length unanswerable examples, but the resulting $p(\mathrm{DK})$ was still not sharp enough to replace confidence thresholding.

The truncation-stage ordering should be understood as part of the progressive-reveal design. We did not keep this order because of uncontrolled data-order effects. Instead, we kept it because the truncated variants are related versions of the same original question. Grouping them preserves the intended curriculum from less information to more information, which appears to help the model learn the reveal pattern. Future work should compare this grouped reveal curriculum against other explicit curriculum designs and multi-seed variants.

Small-scale robustness check. Because repeating the full 100,000-example run for multiple seeds was computationally expensive, we also ran two smaller 10,000-example variants with different seeds. These are not full replicas of the main experiment, but they provide a useful robustness check. In both smaller runs, the maximum-softmax baseline remained stronger near 70% coverage, while the DK model still showed the progressive-reveal pattern: average $p(\mathrm{DK})$ decreased and coverage increased as more of the question became visible. Thus, the reveal-dependent abstention signal is not unique to one run, but it is still weaker than standard confidence thresholding.

The cross-dataset results show domain sensitivity. SciQ is the strongest positive transfer case in absolute terms, although the baseline is still stronger. GSM8K remains a negative result because all methods are below random four-choice accuracy under our artificial conversion. MMLU-Pro shows useful low-coverage gains for DK-based scores, but not a stable advantage across the full curve, and the ten-to-four-choice conversion limits the claim.

7 Conclusion

This project asked whether a small language model can learn to abstain with an explicit DK option when only partial question information is available. Truncation produced the intended qualitative reveal pattern, but by itself risked a length/fragmentation shortcut. Chimera made the signal more principled by adding full-length unanswerable examples. The resulting model learned meaningful abstention behavior, especially on SciQ, but it did not consistently beat maximum-softmax confidence on the main MMLU-style evaluation. The honest conclusion is that Chimera is a promising proof of concept, not a finished solution. Future work should improve answerability labels, for example by using a stronger teacher model to construct higher-quality chimera examples, test stronger curriculum designs, evaluate multiple seeds, and compare against stronger selective-prediction baselines.

Limitations

The experiments use a small model and one main 100,000-example training run. We also ran two smaller 10,000-example variants with different random seeds, which supported the same qualitative conclusions, but these should be treated only as small-scale robustness checks rather than a full multi-seed evaluation. A complete study would repeat the full training pipeline across several seeds and report mean and standard deviation.

GSM8K and MMLU-Pro are converted into four-choice format, which changes the original tasks. Therefore, these results should be interpreted as transfer stress tests for the abstention mechanism rather than standard benchmark results. The no-shuffle choice was used only in the truncation-training stage to preserve the progressive-reveal order of related variants; it should be viewed as a simple curriculum design rather than arbitrary dependence on shuffled order. Finally, the Chimera construction may not perfectly model real unanswerability, because corrupted answer options are only an approximation of cases where a question genuinely lacks enough information to answer.

AI Usage Disclosure and Reflection

Generative AI tools were used to help write and debug notebooks, design evaluation plots, interpret experimental curves, and draft this report. The final claims in the report were checked against the actual experimental outputs and plots. AI assistance was especially useful for organizing the report and identifying missing baselines, but it also created a risk of overclaiming. For that reason, this report explicitly states negative results: the Chimera method does not consistently outperform the baseline, and the cross-dataset transfer is weak outside SciQ.

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